

9. (2.5 points)

An insurance company specializes in coverage for Workers Compensation, Commercial Auto and General Liability. A new actuary is hired to build a reserving model.

- The model uses both internal and external variables.
- The chosen predictors are different than those selected in a previous model, but have a high correlation with claims experience.
- The model was checked for reasonableness by performing diagnostic tests on valuation outcomes.
- Data has been reconciled against the general ledger and the prior analysis.
- The actuary is not yet familiar with past processes and has not yet met with claims staff.
- The actuary models claims from all lines together.

a. (0.75 point)

Briefly describe three main sources of internal systemic risk according to Marshall.

b. (0.75 point)

Identify a reason why this company would score high in a balanced scorecard approach for each of the three sources of risks identified in part a. above.

c. (0.75 point)

Identify a reason why this company would score low in a balanced scorecard approach for each of the three sources of risks identified in part a. above.

d. (0.25 point)

Briefly describe why a portfolio with the risk indicators identified in part b. above would have a lower Coefficient of Variation (CoV) than those identified in part c. above.

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## SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

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| <b>QUESTION 9</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                  |
| <b>TOTAL POINT VALUE: 2.5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>LEARNING OBJECTIVE(S): A5</b> |
| <b>SAMPLE ANSWERS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                  |
| <b>Part a: 0.75 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |
| <p><u>Sample 1</u></p> <ul style="list-style-type: none"> <li>• Specification Error: Error that arises because model is unable to perfectly model the insurance process.</li> <li>• Parameter Selection Error: Error that arises because model is unable to adequately measure all predictors of future claim costs or trends in these predictors.</li> <li>• Data Error: Error that arises due to lack of credible data or lack of understanding or knowledge about the book of business.</li> </ul> <p><u>Sample 2</u></p> <ul style="list-style-type: none"> <li>• Specification error: unable to perfectly model underlying insurance process.</li> <li>• Parameter selection error: unable to select correct parameters &amp; trends.</li> <li>• Data error: lack of data or lack knowledge.</li> </ul> <p><u>Sample 3</u></p> <ul style="list-style-type: none"> <li>• Specification error: the risk that the selected model cannot accurately estimate losses generated by the loss process.</li> <li>• Parameter error: the risk that the best parameters are not used or have not been identified.</li> <li>• Data error: the risk that an effective model can't be fitted because the data is of poor quality.</li> </ul> |                                  |
| <b>Part b: 0.75 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                  |
| <p><u>Sample 1</u></p> <ul style="list-style-type: none"> <li>• Specification Error: The model was checked for reasonableness by performing diagnostic tests.</li> <li>• Parameter Selection Error: Since the model captures both internal &amp; external variables, it will better capture all predictors of future claim cost.</li> <li>• Data Error: Since the data is reconciled to general ledger &amp; prior data analysis, it is reliable &amp; consistent with prior analysis.</li> </ul> <p><u>Sample 2</u></p> <ul style="list-style-type: none"> <li>• Specification: model was checked for reasonableness by performing diagnostic tests on outcomes.</li> <li>• Parameter selection: chosen predictors have high correlation with claims experience. Suggests best predictors selected.</li> <li>• Data error: data reconciled against ledger &amp; prior analysis.</li> </ul> <p><u>Sample 3</u></p> <ul style="list-style-type: none"> <li>• Specification error: diagnostic tests on valuation results.</li> <li>• Parameter error: use internal &amp; external variables.</li> <li>• Data error: data reconciled.</li> </ul>                                                                                       |                                  |

## SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

### Part c: 0.75 point

#### Sample 1

- Specification Error: Since a single model is used for all LOBs the specific patterns unique to each LOB will not be captured.
- Parameter Selection Error: Since predictors are different from prior valuations, they are not consistent over time.
- Data Error: Since the actuary is not familiar with past processes & hasn't spoken to claims he/she may not be aware of any significant episodes or systematic changes & thus may not use the data correctly.

#### Sample 2

- Specification → Actuary models all lines together.
- Parameter Selection → Predictors are different than those used in past analysis.
- Data → Actuary is not familiar w/ past processes & has not met with claims staff.

#### Sample 3

- Specification: Modeling all LOBs together.
- Parameter: Predictors have changed.
- Data: Actuary is not particularly knowledgeable with the process.

### Part d: 0.25 point

#### Sample Responses

- Since the risk indicators in (b) follow best practices the magnitude of risk is reduced. Thus variability in predicted losses is reduced which leads to lower CoV, as compared to risk indicators in part (c).
- The portfolio with good scores will be mapped to a lower CV. The portfolio with bad scores will be mapped to a higher CV.
- Higher score (in part b) means lower volatility → it results in lower CoV.

### EXAMINER'S REPORT

Candidates were expected to briefly describe the three sources of internal systemic risk according to Marshall, identify a high scoring and low scoring characteristic from the listed process, as well as describe how scores in the balanced scorecard approach impact CoV.

#### Part a

Candidates were expected to briefly describe all three sources of internal systemic risk.

Common mistakes included:

- Describing specification error as the risk due to selecting the wrong model or distribution without mentioning the insurance process.
- Describing parameter selection error as the risk of selecting the incorrect parameter values or not outputting the true values of the parameters.
- Listing risk other than internal systemic risks from Marshall.
- Listing the risk without a brief description.

#### Part b

Candidates were expected to identify a high scoring characteristic of the described modeling process in the question.

## SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Common mistakes included:

- Listing high scoring criteria that were not supported by the question.
- Listing low scoring criteria.
- Listing both a high and low scoring criteria such as parameters being different, but highly correlated with claims.

### Part c

Candidates were expected to identify a low scoring characteristic of the described modeling process in the question.

Common mistakes included:

- Listing low scoring criteria that were not supported by the question.
- Listing high scoring criteria.
- Listing both a high and low scoring criteria such as parameters being different, but highly correlated with claims.
- Listing the low score for specification error as only using one model.

### Part d

Candidates were expected to describe why a high scoring portfolio would have a lower CoV than a high scoring portfolio.

Common mistakes included:

- Not describing why variance, or CoV, would be lower for a high scoring portfolio.
- Giving decreased parameter and/or process risk as the reason for lower CoV.